Superpolynomial sample lower bounds for globally margin-conditioned Bayesian neural networks below the mean-field scaling

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Abstract

We study fully connected Bayesian tanh networks on the Boolean cube, conditioned before training to have a fixed absolute output margin at every input. Throughout the positive subcritical scaling regime $0 < \gamma < 1$, we prove a superpolynomial sample lower bound for learning parity. At any fixed depth, after taking the width limit, at least $\exp(\Omega(\sqrt{k}\log k))$ examples are necessary to reduce the posterior expected error below $2/5$ on k-bit parity. The same sample lower bound is necessary to give the correct label probability strictly greater than $2/3$ on more than $9/10$ of the inputs. The bounds hold for every training set of the indicated size. The proof combines the actual network's moderate-deviation rate, an explicit spectral estimate for its saturated kernel, and an elementary polynomial-threshold lower bound. It does not replace the posterior by a Gaussian posterior and therefore also covers $\gamma \geq 1/2$, where higher-order corrections can select among tied Gaussian-rate minimizers. An $\exp(\Omega(k))$ bound at fixed positive test error is not established here.

1 Model and main result

Fix a dimension d, depth $L \geq 2$, width N, margin $a > 0$, and $0 < \gamma < 1$. On $\mathcal{X} = \{-1, 1\}^d$, let

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \ (1 \leq s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

All hidden layers have width N. The readout is linear, there are no biases after the augmented input, and the independent centered Gaussian weights have variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \quad \mathrm{Var}(W^s_{ij}) = N^{-\gamma} \ (1 < s < L), \quad \mathrm{Var}(W^L_i) = N^{-\gamma-1}. \tag{1}$$

The prior is conditioned on the all-input event $A_a = \{|z_N(x)| > a \text{ for every } x \in \mathcal{X}\}$. Training then conditions on exact, noiseless signs $\operatorname{sign} z_N(x_i) = f(x_i)$. Equivalently, the posterior is the original weight law conditioned on A_a and these observed signs.

For a sample S, write

$$p_N(x) = \mathbb{P}\{\operatorname{sign} z_N(x) = f(x) \mid A_a, S\}, \qquad R_N(S, f) = 2^{-d} \sum_x [1 - p_N(x)],$$

and use the confidence criterion

$$F_N(S, f) = 2^{-d} \big|\{x : p_N(x) \leq 2/3\}\big|. \tag{2}$$

Thus R_N is the expected error of a posterior sample, and F_N counts an input as unsuccessful unless its correct-label probability is strictly greater than $2/3$. These are different quantities; in particular this paper does not identify either with the error of the sign of the posterior latent mean.

Here is an explicit version of the lower bound. The more general spectral statement is Theorem 4.3 below.

Theorem 1.1 (A sample lower bound throughout $0 < \gamma < 1$)**.** *Let the target be full parity* $f(x) = \prod_{j=1}^{k} x_j$*, with* $d = k$*. Set*

$$c_L = (2/\pi)^{L-1}, \qquad T_k = \lceil (k+1)\log(4(k+1)) \rceil + 2, \qquad r_k = \lfloor \sqrt{k}/16 \rfloor.$$

Suppose $k \geq 4096$ *and*

$$c_L \binom{k}{r_k + 1} \geq 160(k+1), \qquad n \leq \frac{c_L}{2560 T_k} \binom{k}{r_k + 1}. \tag{3}$$

For every noiseless training sample with at most n *distinct inputs, and for every fixed* $0 < \gamma < 1$*,*

$$\liminf_{N\to\infty} R_N(S, f) \geq \frac{2}{5}, \qquad \liminf_{N\to\infty} F_N(S, f) \geq \frac{1}{10}. \tag{4}$$

At each fixed depth, the sample threshold in (3) *is* $\exp(\Omega(\sqrt{k}\log k))$*.*

The sample locations can be arbitrary and can include repetitions. In particular, the conclusion holds for independent uniform training inputs without an exceptional-sample qualification. Every width limit is taken at fixed d, L, γ, a, S; the statement about growth in k concerns these limiting learners. No uniform interchange of the width and dimension limits is asserted.

Parity is a small-circuit target. Appendix A gives a bounded-weight tanh realization at every fixed depth. The lower bound therefore concerns statistical behavior of this specified prior and conditioning rule, rather than the intrinsic information needed to identify a parity function.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$H(t) = \frac{2}{\pi}\arcsin t, \qquad K(x, y) = H^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right). \tag{5}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x)\geq a \ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{6}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1 + \epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N\to\infty} N^{-\gamma} \log \mathbb{P}\{g(x)z_N(x) > a \ \forall x\} = -E_K(g). \tag{7}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1}\sum_i h_i^s(x)h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \le C_\epsilon e^{-c_\epsilon N} \tag{8}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma}Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1 + x \cdot y)/d$. Because $N^{1-\gamma} \to \infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[H\left(\frac{Q(x,y)}{\sqrt{Q(x,x)Q(y,y)}}\right)\right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there. Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (8). The first-layer limit normalizes the diagonal $(d+1)/d$, giving the kernel in (5).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma}Q_N^{L-1}). \tag{9}$$

Couple it to $Y_N = N^{-\gamma/2}K^{1/2}\xi$, where $\xi \sim \mathcal{N}(0, I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (7), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

3 Spectral estimates with explicit dimension dependence

For $U \subseteq [d]$, let $\chi_U(x) = \prod_{j\in U} x_j$. The characters diagonalize K. Write λ_j for the eigenvalue of the normalized operator $M^{-1}K$ on degree-j characters, and let v denote the smallest eigenvalue of the matrix K itself.

Lemma 3.1 (Spectrum and minimum matrix eigenvalue)**.** *Put $D = d + 1$, $c_L = (2/\pi)^{L-1}$, and $T_d = \lceil D\log(4D)\rceil + 2$. Then*

$$\lambda_0 = \lambda_1 > \lambda_2 = \lambda_3 > \cdots > 0, \qquad v = M\lambda_d,$$

with a final unpaired value when d is even. Moreover,

$$\lambda_0 \geq \frac{c_L}{D}, \qquad v \geq \frac{c_L}{4T_d}, \qquad \lambda_j \leq \binom{d}{j}^{-1} \quad (0 \leq j \leq d). \tag{10}$$

Proof. Write $F = H^{\circ(L-1)} = \sum_{m \text{ odd}} c_m t^m$. All these coefficients are positive, sum to one, and satisfy

$$c_1 = c_L, \qquad c_m \geq c_L/m^2 \quad (m \text{ odd}). \tag{11}$$

Indeed, the coefficient of t^{2j+1} in H is $(2/\pi)\binom{2j}{j}/[4^j(2j+1)]$, and $\binom{2j}{j} \geq 4^j/(2j+1)$ because the largest of $2j+1$ binomial coefficients is at least their average. Each additional composition contributes at least its linear coefficient times the previous series.

On the full D-dimensional cube, the degree-r Fourier eigenvalue of $(u \cdot w/D)^m$ is the probability $p_m(r)$ that the simple coordinate-flip walk, started at a fixed vertex, reaches a specified vertex at distance r after m steps. Expanding the power proves this: the surviving terms are precisely those whose coordinates selected an odd number of times are that specified support. The walk recursion yields

$$\begin{aligned} D[p_{m+1}(r) - p_{m+1}(r+2)] = r[p_m(r-1) - p_m(r+1)] \\ + (D-r-2)[p_m(r+1) - p_m(r+3)] \end{aligned}$$

for $0 \leq r \leq D-2$, omitting terms with zero boundary coefficients. Starting at time zero, induction gives monotonicity with distance within each parity class. Thus the full-cube eigenvalues $\mu_r = \sum_{m \text{ odd}} c_m p_m(r)$ vanish for even r and decrease strictly through odd r: the positive term $m = r$ distinguishes r from $r+2$. Restricting the first coordinate to $+1$ gives $\lambda_r = \mu_r + \mu_{r+1}$. This proves the paired ordering and positive definiteness. The linear coefficient gives the stated lower bound on λ_0. The diagonal $K(x,x) = 1$ gives $\sum_j \binom{d}{j}\lambda_j = 1$, proving the last inequality.

To bound the smallest eigenvalue from below, let $m_0 = \lceil D \log(4D) \rceil$. The Fourier expansion of the walk shows that for $m \geq m_0$ and a vertex of matching parity, the constant and all-coordinate characters contribute $2 \cdot 2^{-D}$. The sum of absolute values of the other contributions, divided by 2^{-D}, is at most

$$\sum_{j=1}^{D-1} \binom{D}{j} |1 - 2j/D|^m \leq 2[(1 + e^{-2m/D})^D - 1] < 1.$$

The last inequality follows from $De^{-2m/D} \leq 1/(16D)$ and $(1+t)^D \leq e^{Dt}$. Consequently $p_m(r) \geq 2^{-D}$ whenever m, r are odd and $m \geq m_0$. Using (11) and comparing a decreasing series to its integral gives, for every odd r,

$$\mu_r \geq 2^{-D} c_L \sum_{\substack{m \geq m_0 \\ m \text{ odd}}} \frac{1}{m^2} \geq \frac{2^{-D} c_L}{2(m_0+2)}.$$

Exactly one of $d, d+1$ is odd, so multiplying $\lambda_d = \mu_d + \mu_{d+1}$ by $M = 2^{D-1}$ yields $v \geq c_L/[4(m_0+2)]$. □

4 Low-energy outputs cannot approximate parity too well

We include the elementary polynomial-threshold argument of Aspnes, Beigel, Furst, and Rudich [1]; no threshold-complexity result is assumed beyond the proof below. At zeros a threshold function may use either fixed sign convention.

Lemma 4.1 (Polynomial thresholds and parity)**.** *If P is a real polynomial of degree at most $r < k$ on $\{-1,1\}^k$, then*

$$\mathbb{P}_x\{\operatorname{sign} P(x) \neq \chi_{[k]}(x)\} \geq b_k(r) := 2^{-k} \sum_{j=0}^{\lfloor (k-r-1)/2 \rfloor} \binom{k}{j}. \tag{12}$$

The same bound holds against a parity on k of d coordinates for a polynomial of total degree at most r on d variables. Also

$$b_k(r) \geq \frac{1}{2} - \frac{r+2}{\sqrt{2k}}. \tag{13}$$

Proof. A sufficiently small constant perturbation makes P nonzero at all cube points while preserving the chosen threshold function. Let $t = \lfloor (k-r-1)/2 \rfloor$. If its error set had fewer than $\sum_{j=0}^{t} \binom{k}{j}$ points, the linear space of multilinear polynomials of degree at most t would contain a nonzero polynomial Q vanishing on that set. Then PQ^2 is nonzero as a cube function, has degree at most $r + 2t < k$, and has nonnegative product with full parity everywhere, with a strictly positive product somewhere. Its inner product with parity is positive, contradicting Fourier orthogonality. This proves (12). For a parity on a subset of coordinates, condition on all the remaining coordinates and average the same bound.

For completeness, the largest atom of $\mathrm{Bin}(k, 1/2)$ is at most $\sqrt{2/k}$. One can obtain this from $\binom{2m}{m}/4^m \leq 1/\sqrt{m+1}$, proved by induction using the ratio $(2m+1)/(2m+2)$; odd k follows from the adjacent even case. By symmetry, the binomial tail in (12) differs from $1/2$ by at most $(r+2)/2$ times the largest atom. This proves (13). □

Lemma 4.2 (A cheap compatible output)**.** *For any target and any sample with at most n distinct inputs, the smallest compatible energy satisfies*

$$E_* \leq a^2 \left(\lambda_0^{-1} + 4n/v \right). \tag{14}$$

Proof. Start with the constant vector $a\mathbf{1}$ and subtract $2a$ at each observed negative-label input. The result has absolute margin a everywhere and agrees with every observed label. Write it as $u + w$, with $u = a\mathbf{1}$ and w supported on at most n points. The quadratic norm inequality gives

$$\tfrac{1}{2}(u+w)^T K^{-1}(u+w) \leq u^T K^{-1} u + w^T K^{-1} w \leq a^2/\lambda_0 + 4a^2 n/v.$$

This output is feasible for one compatible closed-margin orthant, which suffices for its rate energy. □

Theorem 4.3 (A spectral sample lower bound)**.** *Let $f = \chi_U$ with $|U| = k$, and let S have at most n distinct inputs. For any integer $0 \leq r < k$, put*

$$\eta = 2\lambda_{r+1} \left(\lambda_0^{-1} + 4n/v \right).$$

Throughout $0 < \gamma < 1$,

$$\liminf_{N\to\infty} R_N(S, f) \geq b_k(r) - \eta, \tag{15}$$

$$\liminf_{N\to\infty} F_N(S, f) \geq \max\{0, [3b_k(r) - 3\eta - 1]/2\}. \tag{16}$$

In particular, with $D = d + 1$, T_d as in Lemma 3.1,

$$\eta \leq \frac{2[D + 16nT_d]}{c_L \binom{d}{r+1}}. \tag{17}$$

Proof. Take any compatible minimizing sign function g, and let z minimize its closed-margin energy. In the normalized Fourier basis,

$$2E_* = z^T K^{-1} z = \sum_{V \subseteq [d]} \frac{\widehat{z}(V)^2}{\lambda_{|V|}}.$$

Let $P(x) = \sum_{|V| \le r} \widehat{z}(V)\chi_V(x)$. The paired ordering of the eigenvalues and Parseval's identity give

$$\mathbb{E}_x[(z(x) - P(x))^2] \le 2E_* \lambda_{r+1}.$$

Since $|z(x)| \ge a$, a disagreement between $g(x) = \operatorname{sign} z(x)$ and the chosen sign of $P(x)$ implies $|z(x) - P(x)| \ge a$. Markov's inequality and Lemma 4.2 bound the fraction of such disagreements by η. Lemma 4.1 then gives $\operatorname{err}(g, f) \ge b_k(r) - \eta$.

The estimate holds for every compatible minimizing sign, so Proposition 2.1 proves (15) irrespective of the posterior weights among minimizers. At finite width,

$$R_N \le F_N + (1 - F_N)/3 = (1 + 2F_N)/3,$$

which yields (16). Substituting the three spectral bounds from Lemma 3.1 proves (17). □

Proof of Theorem 1.1. Let $d = k$, $r = r_k$, and $B = \binom{k}{r+1}$. Conditions (3) imply

$$k + 1 + 16nT_k \le c_L B/160 + c_L B/160 = c_L B/80.$$

Thus $\eta \le 1/40$. As $k \ge 4096$,

$$b_k(r) \ge \frac{1}{2} - \frac{1/16 + 2/64}{\sqrt{2}} > \frac{17}{40}.$$

Theorem 4.3 gives R_N lower limit at least $17/40 - 1/40 = 2/5$, and F_N lower limit at least $1/10$. Finally, $r + 1 = \Theta(\sqrt{k})$ and $\binom{k}{r+1} \ge [k/(r+1)]^{r+1}$, so the bound on n is $\exp(\Omega(\sqrt{k}\log k))$ at fixed L. □

5 What the bound does and does not establish

The result establishes a fixed-positive-error obstruction for the actual globally conditioned Bayesian learner. In particular, it is stronger than a statement that the exact parity function has tiny prior mass or cannot be the most likely exact function before every point is observed. The confidence conclusion concerns the marginal label probabilities in (2).

The exponential-in-k sample question remains open. The present method first upper-bounds the energy of the learner's minimizers and then bounds what any function of that energy can accomplish. Its limitation is concrete: a polynomial with roots between consecutive Hamming levels in a central band of width $O(\sqrt{k})$ has degree $O(\sqrt{k})$, and its sign agrees with parity on a fixed large fraction of the cube. Scaling that polynomial to a global margin gives kernel energy $\exp(O(\sqrt{k}\log k))$. Thus energy alone cannot give an $\exp(\Omega(k))$ necessary scale at fixed error. A stronger sample lower bound would have to use more about the compatible minimum-energy interpolants and the training constraints. Conversely, that approximation does not prove a learning upper bound: its wrong-sign outputs outside the central band can be very large, so correcting observations there need not be cheap.

The regime $\gamma \le 0$ has a different limiting posterior rather than the energy-minimization limit used here. The theorem makes no sample-complexity assertion for that regime.

A A small bounded-weight parity network

For completeness, full parity on k bits has a one-hidden-layer tanh realization of width $k+1$, with weights $O(k \log(k+1))$ and output margin greater than $1/2$. Let

$$t(x) = \frac{k + \sum_i x_i}{2} \in \{0, \dots, k\}, \quad y_j = (-1)^{k-j}, \quad A = \log(16(k+1)).$$

The step functions $s_j(t) = \operatorname{sign}(t - j + 1/2)$ satisfy

$$y_t = c + \sum_{j=1}^{k} y_j s_j(t), \qquad c = (y_0 + y_k)/2 \in \{0, 1\}.$$

Replace each step by $h_j = \tanh(A(t - j + 1/2))$. Since its argument has absolute value at least $A/2$, its error is at most $2e^{-A}$. A constant neuron $h_0 = \tanh 1$ supplies the term c using output weight $c/\tanh 1$. The resulting output differs from parity by less than $1/8$. First-layer input weights are $A/2$ and bias weights have magnitude at most $Ak/2$; readout weights have magnitude at most 2.

At depth $L \geq 3$, the next hidden layer sums these features and applies tanh, producing the correct sign with magnitude at least $\tanh(7/8) > 1/2$. Each additional hidden layer can preserve this margin by a single weight 2, since $\tanh(2u) > 1/2$ whenever $u > 1/2$. A final readout weight 1 completes the construction. Unused neurons are set to zero. Thus the same polynomial bounds hold at every fixed depth.

References

[1] J. Aspnes, R. Beigel, M. Furst, and S. Rudich. *The expressive power of voting polynomials.* Combinatorica 14(2):135–148, 1994. Author-hosted manuscript.